

EVOLUTIONARY BIOLOGY

The convergent evolution of the pelvic suction cup: A unique key innovation in spiny-finned fishes

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Key innovations open ecological opportunities and can redirect evolutionary trajectories. In spiny-finned fishes, the pelvic suction cup—a fused adhesive structure formed from the pelvic fins and supported by a modified girdle—appears to be one such trait. Using a novel 960-species phylogeny spanning 940 exons and 67 newly sequenced snailfish genomes, we show that the suction cup evolved independently three times across lineages separated by over 100 million years. By enabling adhesion in high-energy habitats such as tide pools and waterfalls, the suction cup set the stage for body depression and accompanying shifts in scales, teeth, and clade-specific ecofunctional profiles. Comparative analyses reveal convergence in a distinctive region of morphospace and body-shape evolution accelerated two- to fivefold. Diversification analyses uncover heterogeneous but elevated rates, with a clear burst in rock-climbing gobies associated with suction cup evolution. By opening novel habitats and fostering phenotypic novelty, the suction cup emerges as a key innovation that reshaped spiny-finned fish evolution.

INTRODUCTION

One of the most enduring puzzles in evolutionary biology is why morphological diversity and diversification rates are so unevenly distributed across the tree of life (1–3). A leading hypothesis is that key innovations—evolutionary novelties that unlock ecological opportunities—can trigger ecological shifts that, in turn, drive elevated rates of lineage diversification or accelerated morphological evolution. The term “key innovation” was first introduced by A. H. Miller in 1949 to describe “key adjustments in morphological and physiological mechanisms” that were essential for the origin of major clades (4). Over time, this concept has been refined to focus less on the origin of higher taxa and more on novel traits that promote diversification within a clade (5–7). The intuitive appeal of key innovations is bolstered by conspicuous and charismatic examples, including the autopod in tetrapod limbs (8), wings for powered flight in bats (9, 10), or feathers in birds (11, 12), and bipedalism in hominids (13, 14). The key innovation framework, however, has also faced considerable scrutiny: Critics have argued that early studies suffered from confirmation bias, simply finding synapomorphies of highly diverse clades

(10, 15). In response, modern macroevolutionary studies have sought to identify key innovations using explicit criteria. Traits proposed as key innovations are expected to enable access to previously inaccessible ecological opportunities, broadly defined as habitats, resources, or other parts of the environment that could not previously be used (7). They are often characterized by one or more of the following signatures: evolution of new morphologies in response to the new adaptive landscape (6, 16), accelerated rates of morphological evolution (17, 18), and bursts of increased lineage diversification (15, 19). Although meeting all criteria is not strictly necessary, the most compelling cases tend to satisfy them all (16, 20–22).

Spiny-finned fishes, or acanthomorphs, account for nearly a third of all extant vertebrate species and represent a wide variety of body plans (23–25). However, diversity is not evenly distributed across subclades. There have been multiple independent radiations in a variety of phenotypic trajectories without any single environmental or geographic factor able to explain all radiations, suggesting a suite of unique innovations (26–30). From highly protrusible jaws to anti-freeze proteins, Wainwright and Longo (30) reviewed eight such innovations that have been instrumental in the adaptive radiation of this group. Strikingly, none of these involve paired fins despite the emphasis on paired limb modifications in tetrapod research (31–33). Yet paired fins in bony fishes also display a broad range of specialized modifications. For example, flying fishes have greatly enlarged, wing-like pectoral fins that enable gliding above the water’s surface (34), and amphibious mudskippers have pectoral fins specifically modified for terrestrial “walking” on land (35).

One such paired fin modification in bony fishes is the pelvic suction cup, exemplified by rock-climbing gobies, clingfishes, and lumpsuckers/snailfishes. We define the suction cup as the presence of both form and function: A transformed pelvic girdle supports fused pelvic fins that form a cup, together allowing fishes to create and sustain strong suction forces (36, 37). The pelvic suction cup has been examined extensively from morphological and biomechanical perspectives (36–42), but its downstream impact on evolutionary trajectories has received far less attention. This is unexpected because the suction cup can

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enable access to ecological opportunities that were previously inaccessible, which Miller *et al.* (7) define as “[access to] habitats, resources, or other parts of the environment that previously could not be used.” Rock-climbing gobies are able to scale waterfalls to reach new habitats (37); clingfishes, whose adhesive disc is additionally supported by pectoral elements, can generate such powerful adhesive force that they can hold 80 to 250 times their body weight (40, 42) and are the only fishes found in certain high-action tidal habitats (36, 43); and snailfishes are able to withstand high wave action in the shallows and cling to crustaceans in the deep, facilitating reproductive parasitic behavior in which they deposit eggs in crabs’ gills (44, 45). In addition, the suction cup may aid trench-dwelling snailfishes in maintaining position during intermittent turbulence events (46). Previous research shows that rock-climbing gobies are the most diverse group of tropical freshwater gobies (47), snailfishes exhibit exceptional rates of phenotypic evolution and lineage diversification (29, 48, 49), and clingfishes display remarkable morphological and ecological variability (50). These observations make the pelvic suction cup a clear potential candidate for a key innovation among spiny-finned fishes.

Because the pelvic suction cup has arisen independently in three acanthomorph lineages separated by over 100 million years (MY), each involving reconfiguration of pelvic rays and the pelvic girdle (Fig. 1, A to D), it provides phylogenetic replication for testing key-innovation predictions. Key-innovation theory recognizes that the same functional trait can originate under different selective pressures and shape evolutionary success through different ecological mechanisms (6, 7, 51). We therefore do not assume that these independent origins must produce identical signatures in phenotypic evolution, ecofunctional consequences, or lineage diversification. Instead, we treat them as replicates that allow us to test which macroevolutionary consequences are general and which depend on lineage-specific ecologies and constraints (52).

Together, the repeated origins of the pelvic suction cup across distantly related acanthomorph lineages provide an exceptional opportunity to evaluate both the general evolutionary dynamics of this structure and whether it functions as a key innovation. We expect the following macroevolutionary signatures: (i) asymmetry in evolutionary transitions, with suction cup evolution being effectively irreversible once lost; (ii) consistent correlations between suction cup origin and both discrete and continuous morphological shifts reflecting changes in selective pressures related to the suction cup (e.g., wide and dorsoventrally depressed bodies that maximize adhesive surface area and reduce drag); (iii) occupation of distinctive and expanded regions of morphospace and changes to functional distinctiveness/specialization, potentially reflecting previously unidentified adaptive regimes; (iv) accelerated rates of body-shape evolution in suction cup-bearing lineages; and (v) elevated lineage diversification, either sustained or in bursts, following suction cup origins. Testing these predictions requires both dense phylogenomic sampling and comparative models applied across the three independent evolutionary experiments provided by clingfishes, snailfishes/lumpsuckers, and rock-climbing gobies.

Using a newly constructed 960-species percomorph phylogeny, including 67 newly sequenced snailfish genomes, we combine phylogenetic comparative methods with morphometric and ecological datasets to evaluate these predictions. Specifically, we reconstruct the evolutionary history of pelvic-fin morphologies, test models of transition asymmetry, assess correlations with discrete and continuous

traits, quantify morphospace occupation and ecofunctional distinctiveness or specialization, estimate rates of morphological evolution, and examine diversification dynamics across suction cup-bearing clades. Through these integrative analyses, we aim to assess whether the pelvic suction cup represents a unique, previously overlooked key innovation in spiny-finned fishes.

RESULTS AND DISCUSSION

Large-scale phylogenomic analysis reveals that suction cup loss is common among snailfishes but rare or absent in other clades

We inferred a 960-species percomorph phylogeny using a combination of maximum-likelihood (ML) and coalescent-based approaches, applied to 940 exons from 552 species—including 423 newly sequenced taxa, 67 of which are newly generated snailfish genomes. Taxon sampling emphasized three clades of percomorphs (a large subclade of acanthomorphs) with pelvic suction cups—rock-climbing gobies, clingfishes, and cyclopteroids (lumpsuckers + snailfishes)—along with their close relatives [hereafter “type 1” (T1) outgroups], while including sparser coverage of other percomorph lineages [“type 2” (T2) outgroups] to ensure representation of all major groups. For brevity, we use RCGobies+T1 for rock-climbing gobies and other oxudercid gobies; clingfishes+T1 for clingfishes, blennies, jawfishes, and basslets; and cyclopteroids+T1 for snailfishes, lumpsuckers, sculpins, poachers, and eelpouts. To address phylogenetic uncertainty, we generated 10 alternative trees using genomic subsets and then expanded the taxon sampling to 960 taxa by adding one aligned mitochondrial and eight nuclear legacy markers (table S1) from GenBank to each concatenated matrix (see Supplementary Data for accession numbers). In addition, for snailfishes, we constructed the most comprehensive phylogeny to date (143 species and 2900 exons) by integrating BUSCO sequences (53) from newly sequenced genomes and legacy markers. We time-calibrated all trees (10 full percomorph backbones and 2 snailfish phylogenies) via congruification using 246 secondary calibrations from reference time trees (27, 49, 54–57) and penalized likelihood, and subsequently grafted snailfish trees onto the full percomorph backbone to yield the final 960-species phylogenies (fig. S1).

Phylogenetic analyses and ancestral state reconstructions using Markov (Mk) model averaging (see details in the next section) confirm that the pelvic suction cup originated independently three times, across three major percomorph clades (Ovalentaria, Gobiaria, and Eupercaria) separated by over 100 MY (Fig. 2). To assess the robustness of the Mk model-based estimates, we additionally estimated gains of the suction cup using a parsimony-based reconstruction, which does not assume stationarity and performs well when the rate of evolution is slow. These reconstructions likewise recovered three independent origins (see table S4). On the basis of our time calibrations, the suction cup emerged in the Cenozoic: once in the late Eocene (clingfishes; 38.8 MY), once in the late Oligocene (lumpsuckers and snailfishes; 25.5 MY), and once in the mid-Miocene (rock-climbing gobies; 11.2 MY). These phylogenies also revealed that losses of the suction cup were rare in two clades, with no losses of pelvic elements in rock-climbing gobies and only one in clingfishes (genus *Alabes*). In contrast, snailfishes lost all external pelvic structures independently at least seven times (Fig. 2 and fig. S2). The drivers of frequent suction cup loss in snailfishes are unclear: Depth alone is unlikely given that the deepest dwelling species retain the

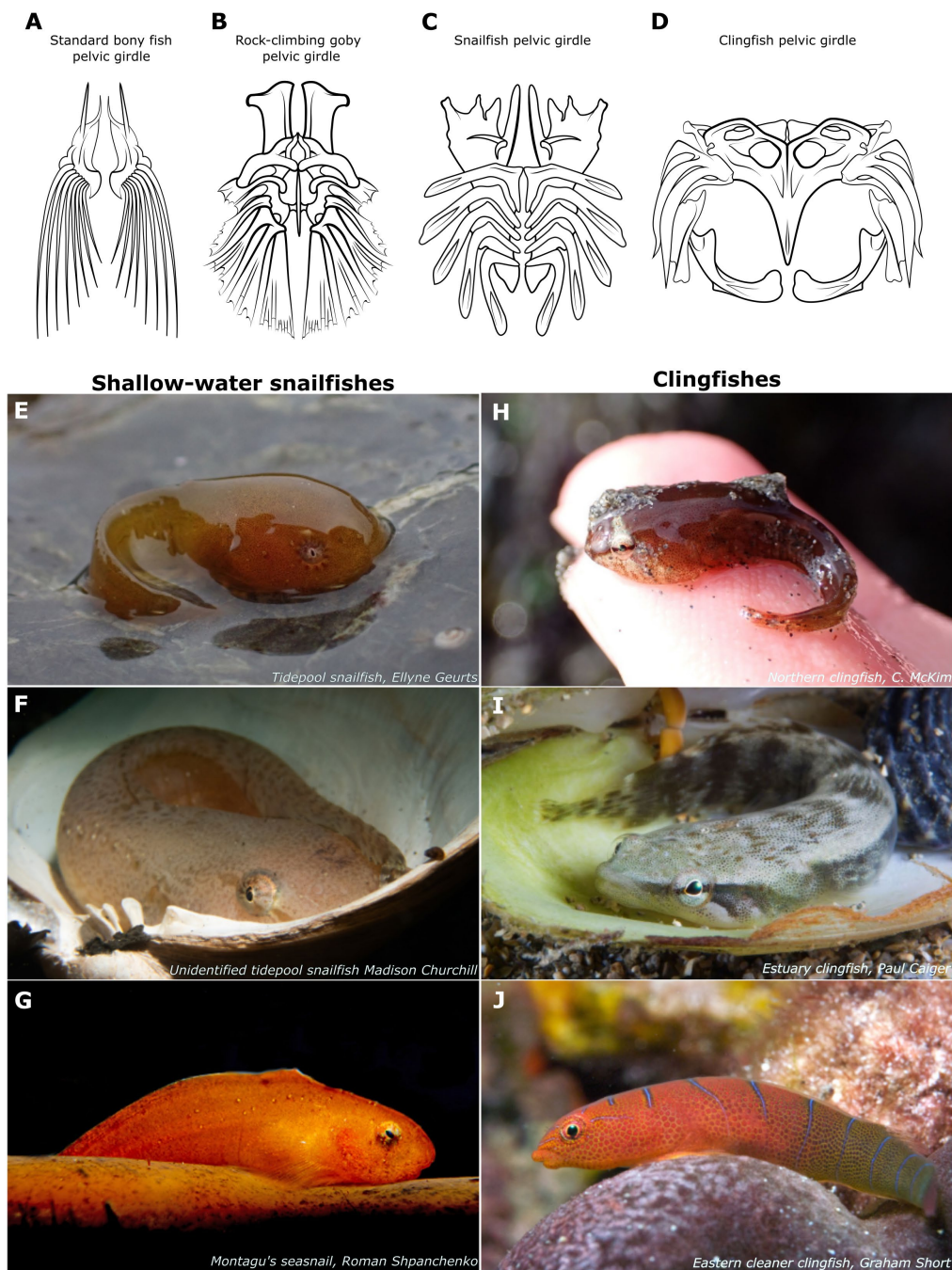


Fig. 1. The skeletal pelvic morphology of the pelvic suction cup and convergence in shallow-water suction cup-bearing fishes. Pelvic girdle skeletons of (A) a typical ray-finned fish with separate pelvic fins [*Barbus intermedius*, modified from (136)]; (B) a rock climbing goby [*Awaous stamineus*, adapted from (37)]; (C) a snailfish [*Liparis dennyi*, adapted from (36)]; and (D) a clingfish [*Gobiesox maeandricus*, adapted from (36)]; (disc supported by modified pelvic and pectoral elements; pectoral components not shown). Images show examples of notable body shape convergence between snailfishes (E to G) and tidepool clingfishes (H to J), two lineages separated by ~100 MY of evolution yet often mistaken for one another on iNaturalist. (E) *Liparis florae*, the tidepool snailfish, by E. Geurts/public domain. (F) *Liparis* sp., by M. Churchill under a CC-BY-NC license (<https://creativecommons.org/licenses/by-nc/4.0/deed.en>). (G) *Liparis montagui*, Montagu's sea snail, photographed by R. Shpanchenko/used with permission by photographer. (H) *G. maeandricus*, the northern clingfish, by C. McKim under a CC-BY-NC license (<https://creativecommons.org/licenses/by-nc/4.0/deed.en>). (I) *Trachelocheismus aestuarium*, the estuary clingfish, by P. Caiger/used with permission by photographer. (J) *Cochleocephalus orientalis*, the eastern cleaner clingfish, by G. Short/used with permission by photographer.

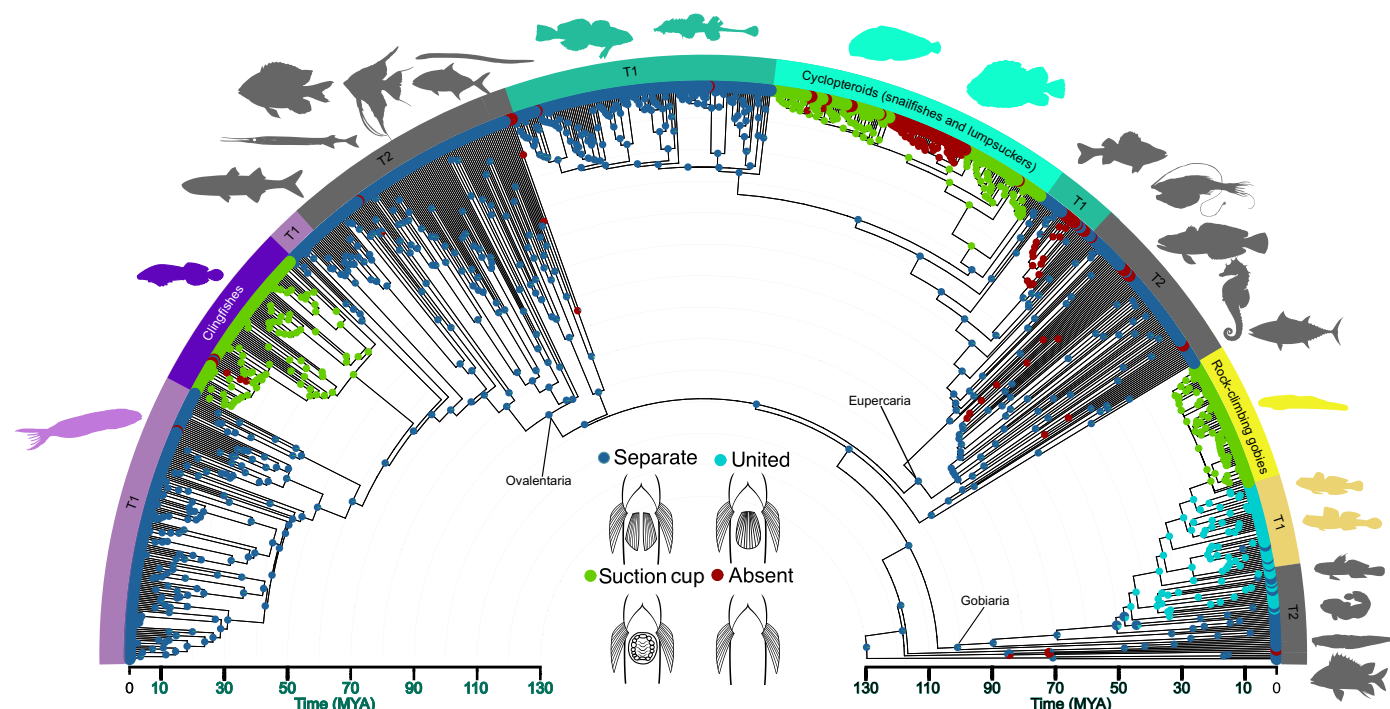


Fig. 2. The pelvic suction cup has originated three separate times across spiny-finned fishes. Model-averaged ancestral state reconstruction of pelvic-fin morphology for 960 species reveals three origins of the pelvic suction cup. Focal groups have been highlighted in color and labeled, with nonfocal groups in gray, labeled as T2 (type 2 outgroups). Groups labeled T1 (type 1 outgroups) are the closest relatives of the suction cup-bearing clades that were densely sampled to avoid taxonomic bias in downstream analyses. Losses of the suction cup are common among snailfishes with no regains, in contrast to frequent regains of separate pelvic fins. The inset diagrams are ventral views of the pelvic-girdle region.

suction cup (39). Losses may instead reflect interactions between depth and life history, given larval differences between groups: Deep-living species with the suction cup have demersal larvae, contrasting the pelagic larvae of shallow-living cup-bearing species and deep-living species without suction cups, a pattern noted by J. W. Orr from extensive observations across snailfish taxa. The estimate of losses among snailfishes is likely an underestimate given the absence of molecular data for more than 200 species, many of which are difficult to sample. Even so, these estimates rank snailfishes among the vertebrate families with the highest number of repeated pelvic appendage losses, rivaling skinks (58) and sticklebacks (59). This robust, expansive phylogeny provides a foundation for macroevolutionary inferences, as detailed below.

Transitions between pelvic-fin phenotypes are asymmetric and directional; suction cup evolution is irreversible

To investigate how pelvic-fin phenotypes evolved across percomorphs, we fit models of discrete character evolution to all 10 percomorph trees using the R package *phytools* (60). Pelvic fins were coded into four discrete states: separate (ancestral condition), united (membrane-connected but lacking skeletal modifications), suction cup (external suction cup with underlying skeletal restructuring; see Materials and Methods for considerations on coding this character), and absent. We fit models to the complete tree to examine clade-specific loss dynamics and reversibility of transitions within the three subclades containing lineages that have the suction cup (RCGobies+T1, clingfishes+T1, and cyclopteroids+T1). We compared standard Mk models [equal rates (ER), symmetrical (SYM), and all-rates-different

(ARD)] and their variants that treat the character as a threshold trait (60), along with four custom models incorporating biologically informed constraints. These custom models varied in the directionality and reversibility of transitions, including whether lineages were required to pass through an intermediate “united” state and whether fin loss was irreversible. To further validate our rate estimates and assess potential bias from infrequent suction cup gains, we fitted continuous-time Mk models using *corHMM* (61), which allow for rate variation testing an ARD model with zero, one, and two hidden rate categories. The best-supported Mk model across the full tree was the custom unordered, loss-reversible (ULR) model (Fig. 3A), which allows direct transitions from separate fins to the suction cup, permits transitions out of the absent state, but prohibits transitions from the suction cup to the separate state. In *corHMM*, the best-supported model for all 10 trees was the model with no hidden rate categories, and transition rate estimates aligned with those estimated by *fitMk* (table S5).

The pelvic suction cup shows notable evolutionary asymmetry: losses of the suction cup (two → three) are several times more common than gains (Fig. 3A), and reversals back to finned states (two → zero or two → one) were not observed. These Mk-based rate estimates were nearly identical to those obtained from the best-fit model estimated by *corHMM* (see table S5). Ancestral state reconstructions on all full trees recovered only one putative regain (three → two) of a vestigial suction cup in the clingfish genus *Alabes* (62), although this inference depends on uncertain phylogenetic placement (see stacked bar plots in Fig. 3 and Supplementary Data for all tree topologies). By contrast, separate pelvic fins were repeatedly regained across

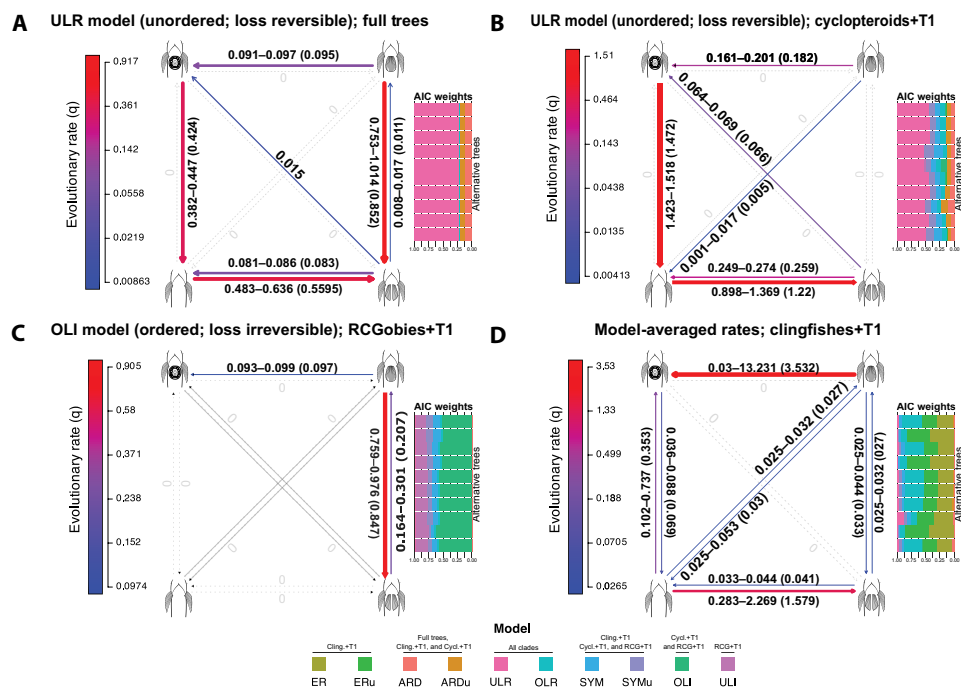


Fig. 3. Transitions between pelvic-fin morphologies across percomorph fishes are asymmetric. (A) The best-fit model (ULR) for full trees allows direct transitions from separate fins to the suction cup, permits regain of pelvic elements after complete loss, and restricts suction cup lineages to pelvic-element loss rather than reversion to separate or united fins. Estimated transition rates show that suction cup losses are more common than gains and that the suction cup is not regained once lost, whereas separate pelvic fins are regained more often than they are lost. (B) Cyclopteroids+T1 also favor ULR, matching the full-tree result. (C) In RCGobies+T1, the best-fit model (OLI) requires a united intermediate and assumes irreversibility of pelvic-element loss. (D) In clingfishes+T1, model support varies across trees with no single best-fit model (fig. S4). Inset stacked bar plots represent AIC weights for each model for each alternative tree topology.

the phylogeny, aligning with previous studies that recovered a minimum of 14 pelvic fin regains among bony fishes (63). Subclade-specific model fits revealed distinct evolutionary trajectories: RCGobies+T1 supported the custom ordered, loss-irreversible (OLI) model, requiring the united state as an intermediate; clingfishes+T1 showed comparable support for ER and its threshold variant; and cyclopteroids+T1 favored the ULR model.

These results indicate that while transition dynamics differ among clades (especially with respect to frequency of pelvic element loss), the transition to a suction cup is broadly constrained, with no credible regains once pelvic elements are lost. Similarly, unlike the frequent reversions from united to separate fins in gobies, the suction cup never reverts to separate fins. These asymmetries are consistent with extensive remodeling required to evolve a suction cup and with stronger constraints governing its evolution.

Evolution of the suction cup is correlated with the presence of complex teeth, loss of scales, and dorsoventral depression

Anatomical literature suggests that the pelvic suction cup is associated with a suite of traits that may either precede or follow ecological shifts enabled by adhesion (62–65). We coded three discrete traits: tooth type (unicuspid versus multicuspid; 538 spp.), scale presence (940 spp.; linked to hydrodynamics/drag), and swim bladder presence (929 spp.; associated with buoyancy). Tooth type was included because multicuspid dentitions are often tied to trophic function and could plausibly covary with substrate use enabled by adhesion (65). We also aggregated a dataset of five continuous, morphometric traits (body elongation, head depth, body depth, body width, and

dorsoventral depression) for 538 species, following the FishShapes database, combining newly collected measurements from 429 specimens with 1070 FishShapes records (66). We tested for associations between pelvic suction cup evolution and discrete/continuous traits using corHMM correlation models and phylogenetic logistic/linear regression in Phylolm (67) and assessed significance with false discovery rate (FDR)-corrected *P* values (see Materials and Methods and Supplementary Data for trait definitions and dataset).

Across the full percomorph phylogeny, corHMM generally favored hidden-rate models, indicating substantial background rate heterogeneity (table S6). Evidence for correlated evolution was then evaluated by comparing dependent-hidden models against independent-hidden nulls (rate heterogeneity unrelated to the focal traits; table S6), and support for dependence was trait-pair specific. Phylolm detected no association between the suction cup and swim bladder or scale absence but supported a relationship with multicuspid teeth when the suction cup was the predictor ($P = 2.61$ to 5.61×10^{-16} across the 10 alternative trees) and, more weakly, as the response ($P = 0.022$ to 0.047). As sensitivity analyses, we also ran the same framework within each focal subclade (table S6), and patterns were lineage dependent: RCGobies+T1 showed no relationships after FDR correction; cyclopteroids+T1 supported multicuspid teeth as both predictor ($P = 4.69$ to 5.02×10^{-5}) and response ($P = 0.006$ to 0.013); and clingfishes+T1 supported associations when swim bladder absence ($P = 0.002$ to 0.013) and scale absence ($P = 0.003$ to 0.019) were treated as predictors, and in all trees when multicuspid teeth were treated as the response ($P = 1.29 \times 10^{-7}$ to 0.003). Overall, the most repeatable signal links the suction cup to multicuspid teeth (see below

for possible ecofunctional correlates), whereas associations with scales and swim bladder loss are weak or clade specific.

Suction cup-bearing fishes also exhibited a characteristic body plan. Phylogenetic generalized linear models (67) showed that suction cup-bearing species are more dorsoventrally depressed, less elongate, and have wider bodies and more depressed heads than their relatives (all $P < 0.01$; Fig. 4A and table S7), with shifts strongest in

clingfishes, moderate in cyclopteroids, and absent in rock-climbing gobies. Models in which changes to continuous characters are predicted by pelvic fin-state received the strongest support, whereas the inverse models failed to reach significance and/or received weaker AIC support.

Our results indicate that the pelvic suction cup is a strong predictor of consistent shifts in body shape in the context of the full

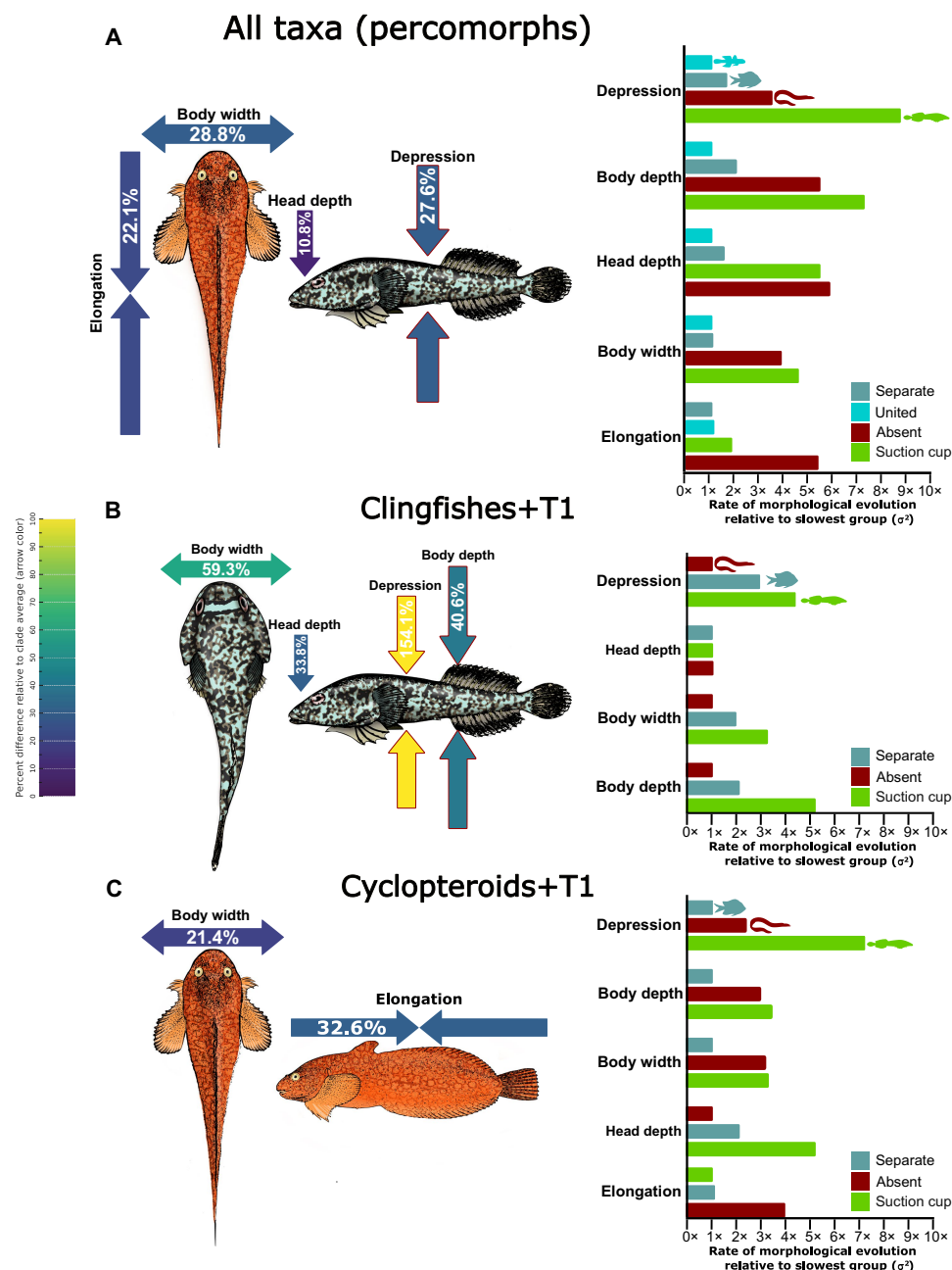


Fig. 4. Trends in body shape associated with the evolution of the pelvic suction cup across spiny-finned fishes. (A) Phylolm analyses reveal that percomorph species with a pelvic suction cup are, on average, 27.6% more dorsoventrally depressed, 28.8% wider, 22.1% less elongate, and have a head that is 10.8% shallower than their cupless relatives. The best-fit model (multiregime, multi-optima, fixed alpha Ornstein-Uhlenbeck, OUMV) estimates that these traits evolve at 2.5 to 8.0x (body depression), 1.3 to 6.6x (body depth), 1.2 to 4.2x (body width), 0.3 to 1.7x (elongation), and 0.9 to 5x (head depth) the rates observed in cupless lineages. Clade-specific results for (B) clingfishes+T1 and (C) for cyclopteroids+T1 (snailfishes, lumpfishes, and close relatives). Rock-climbing gobies are not depicted because analyses revealed no significant differences between suction cup-bearing RCGobies and their T1 outgroups.

percomorph phylogeny, with clade-specific differences in magnitude. Although we cannot resolve the temporal sequence of these changes along the corresponding stem lineages, the directional pattern is consistent with an evolutionary scenario in which attachment to high-energy benthic habitats favors a low-profile, depressed body plan and associated functional shifts (68–71) with possible relaxation of selection on swimming-related features such as scales and swim bladders (72). This mirrors one of two major types of innovations that influence morphological diversity, as defined by Wainwright (73): traits that change the adaptive landscape an organism is exposed to.

Morphological and functional consequences of the pelvic suction cup

Accelerated morphological evolution is increasingly recognized as a robust signature of novel adaptive traits (22, 74–78). For example, Rabosky *et al.* (48) showed that snailfishes have the second fastest rates of both body-size evolution and lineage diversification among ray-finned fish families, and benthic lineages more broadly exhibit greater body-shape diversity and faster phenotypic evolution than their pelagic or demersal counterparts (79). Clingfishes, benthic specialists marked by extreme body forms and ecological diversity (50), are another potential example. We therefore predicted that suction cup-bearing clingfishes and snailfishes would exhibit elevated rates of morphological evolution linked to their distinctive ecology and anatomy. Using our comprehensive percomorph phylogeny and continuous-trait dataset, we tested whether patterns of morphospace exploration in these groups can be attributed to the pelvic suction cup by comparing evolutionary models and parameter estimates across the full tree and within focal clades. A consistent pattern of greater trait divergence in suction cup-bearing fishes would support the hypothesis that the suction cup functions as a key innovation.

Using our morphometric dataset, we fit single- and multiregime univariate Brownian Motion (BM1 and BMS) and Ornstein-Uhlenbeck models (OU1, OUM, OUMV, OUMA, and OUMVA) in OUwie (80) for each continuous variable, coding pelvic-fin morphology as “separate,” “united,” “suction cup,” or “absent.” Variable-alpha OU models (OUMA and OUMVA) were excluded from further analyses because of unrealistically high log-likelihood values and/or negative eigenvalues, consistent with known instability from overparameterization and confounding between alpha and σ^2 (80, 81). In addition, fixing alpha across regimes (while allowing σ^2 to vary, as in OUMV) reduces confounding of parameters and allows interpretation of regime-specific σ^2 as relative stochastic divergence under a shared reversion timescale (82). Across the full tree, OUMV was favored (table S8), and suction cup-bearing species showed the highest σ^2 values for several traits, including body depression ($\sigma^2 = 0.319$ versus 0.040 to 0.129 in other states), body width ($\sigma^2 = 0.126$ versus 0.030 to 0.107), and body depth ($\sigma^2 = 0.073$ versus 0.011 to 0.055; Fig. 4A), whereas species lacking pelvic fins had the highest σ^2 values for head depth ($\sigma^2 = 0.118$, only marginally higher than the suction cup: $\sigma^2 = 0.110$) and elongation ($\sigma^2 = 0.202$; Fig. 4A and table S9). Estimated trait optima (θ) for fishes with a suction cup are consistent with a wide, somewhat stout, dorsoventrally depressed body (table S10). These results indicate that suction cup-bearing lineages show elevated stochastic divergence in body shape, with inferred optima consistent with a wide, dorsoventrally depressed body plan.

Analyses of individual subclades were consistent with this pattern. In gobies (RCGobies+T1), OU1 best fit the data, consistent with

overlapping morphospaces and limited phenotypic divergence. This pattern is consistent with prior work suggesting an inverse link between species richness and phenotypic diversity in gobies and cardinalfishes (83), although these studies emphasized disparity rather than rates (84–88). This may reflect constraint on a generalized goby body shape or preadaptation for fast-flowing streams. In clingfishes+T1, OUMV was favored, with suction cup-bearing species showing the highest σ^2 for body depression ($\sigma^2 = 0.52$) and body depth (0.05), whereas finless species showed the highest σ^2 for elongation (0.37; Fig. 4B). Estimated optima (θ) for suction cup-bearing fishes were consistent with the expected wide, dorsoventrally depressed, stout body plan (see table S10 for estimated θ values). In cyclopteroids, BMS was generally favored, with suction cup-bearing species showing elevated rates of trait evolution under BM (for example, depression: 0.37 versus 0.07 to 0.15; width: 0.09 versus 0.03 to 0.07; Fig. 4C and table S8). Overall, suction cup-bearing lineages tend to show two- to threefold higher BM rates and/or higher OU stochastic variance than other pelvic-fin morphotypes, depending on the best-supported model, consistent with a key-innovation interpretation.

Next, we asked whether these morphological patterns are accompanied by shifts in ecofunctional profiles (functional distinctiveness/specialization) relative to T1 outgroups. To test for corresponding shifts in ecological function, we quantified ecofunctional profiles from clade-specific ecological trait data (see Materials and Methods and table S11) using the *gowdis* function in the R package FD (89) and a principal coordinate analysis (PCoA) with a Lingoes correction, and then compared suction cup-bearing taxa to their closest outgroups. These analyses revealed clade-specific ecological consequences: Relative to their T1 outgroups, rock-climbing gobies showed lower functional distinctiveness [$P_{\text{FDR}} = 0.0001$, coefficient of determination ($R^2 = 0.87$; Fig. 5F)], consistent with reduced ecofunctional differentiation associated with specialization in fast-flowing stream habitats and algae-scraping (47), whereas clingfishes exhibited marginal functional expansion ($P_{\text{FDR}} = 0.051$, $R^2 = 0.87$; Fig. 5G), indicative of their diverse microhabitat affinities. Despite their different morphological regimes, no shift in ecological and functional space occupancy for cyclopteroids was detected relative to outgroups (Fig. 5H), even given the breadth and global distribution of snailfishes (39, 63). See also figs. S6 and S7.

Last, to elucidate potential evolutionary correlations between tooth type (i.e., unicuspid or multicuspid) and its relationship to ecological traits, we performed phylogenetic generalized logistic regression analyses in R package Phylolm (67), both in univariate and multivariate frameworks on all three taxon sets, testing correlations with all ecological variables examined. These analyses supported an association between trophic category and tooth type within RCGobies+T1, with multicuspid teeth linked to herbivory ($P_{\text{FDR}} = 0.003$ to 0.005). A second signal involved broad salinity use ($P_{\text{FDR}} = 0.003$ to 0.08), which in this clade is tightly coupled to amphidromy; we therefore interpret diet as the more direct functional correlate. No significant outcomes were produced within clingfishes+T1 or cyclopteroids+T1 (table S12).

These results show that the evolutionary consequences of the suction cup are clade specific. Across the full tree, suction cup-bearing lineages show elevated phenotypic divergence in body shape, expressed as higher stochastic divergence under the best-supported OU model (OUMV) for several traits and inferred optima consistent with a wide, dorsoventrally depressed body plan (table S10). Subclade analyses indicate that this signal is clade dependent: Cyclopteroids and

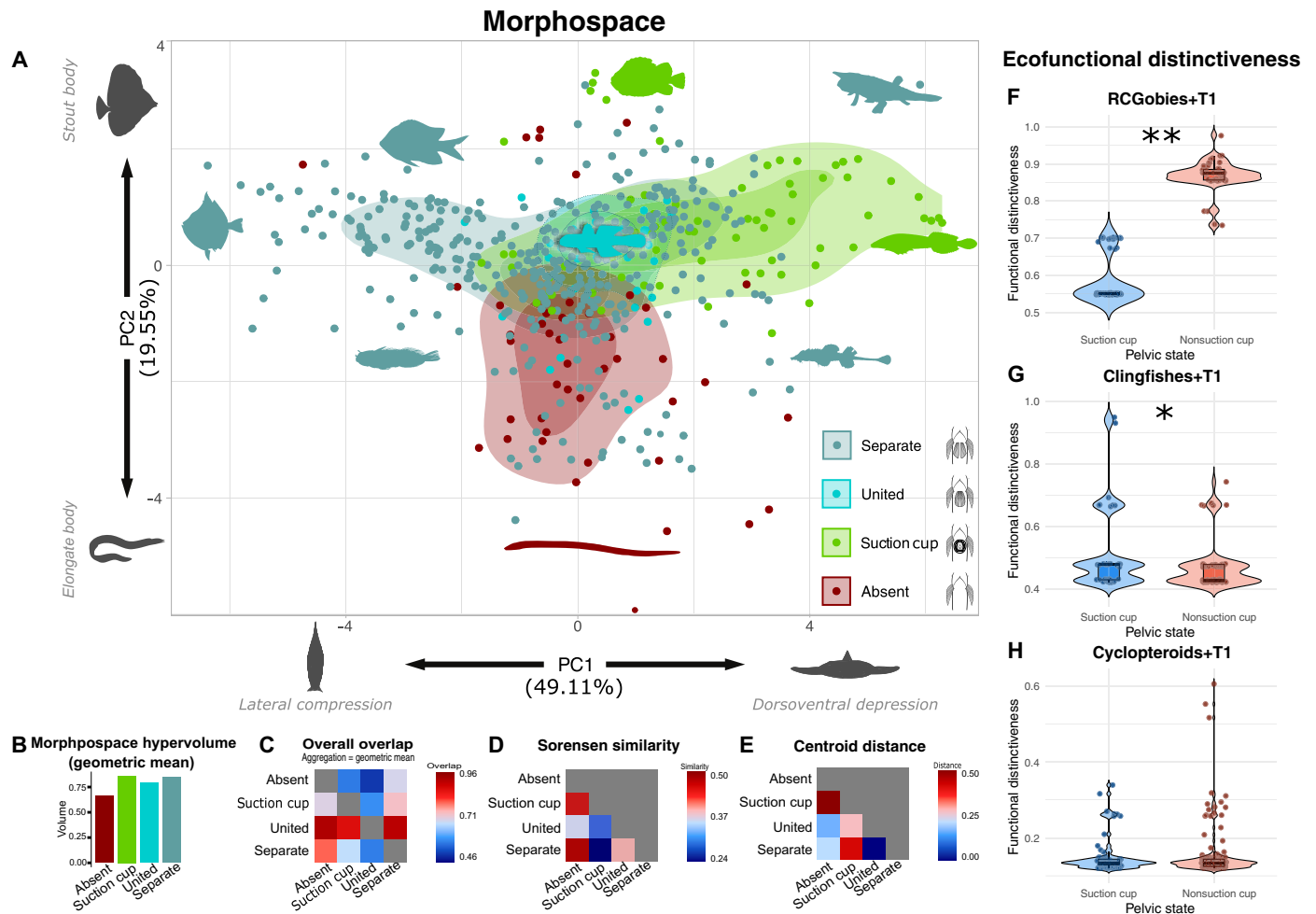


Fig. 5. Hypervolumes, principal component, and functional distinctiveness analyses reveal that suction cup-bearing fishes occupy a broad and distinct area of morphospace and exhibit clade-specific functional shifts. (A) PC1 to PC2 morphospace with points and two-dimensional hypervolume slices colored by pelvic morphology. Darker shading indicates regions where local point density is at least two-thirds of the maximum, and lighter shading indicates at least one-third; dotted outlines highlight the united state. (B) Hypervolume size for each pelvic morphology group (geometric mean). (C) Heatmap of hypervolume overlap (geometric mean); each cell shows the proportion of the vertical-axis hypervolume overlapping the horizontal-axis hypervolume (asymmetric). (D) Heatmap of Sørensen similarity between hypervolumes. (E) Heatmap of centroid distances between hypervolumes. (F to H) Functional distinctiveness comparisons show clade-specific patterns, with the suction cup associated with functional contraction (**) in RCGobies+T1 (F), marginal (*) functional expansion in clingfishes+T1 (G), and no detectable shift in cyclopteroids+T1 (H).

clingfishes show greater variance around optima under OUMV with the latter additionally showing marginal increases in ecofunctional differentiation relative to their outgroups, whereas rock-climbing gobies fit a single-regime model (OU1) with limited phenotypic divergence and show reduced ecofunctional differentiation consistent with specialization in fast-flowing stream habitats. Thus, while the suction cup is associated with elevated phenotypic divergence, its ecological consequences vary among lineages and may be partly obscured by data limitations, particularly in deep-living groups such as snailfishes.

All in all, this combination of patterns meets a key criterion of evolutionary innovation: entry into a previously unexplored adaptive zone followed by evolutionary diversification or specialization, without requiring a uniform ecological response across lineages (15). Suction cup evolution is followed by reduced ecofunctional differentiation in rock-climbing gobies, marginal increases in ecofunctional differentiation in clingfishes, and no detectable functional shift in cyclopteroids at the scale of the ecological trait categories used here.

At the same time, the repeated association between the suction cup and elevated rates of phenotypic evolution supports a key-innovation interpretation since elevated and repeated morphological divergence is widely treated as a hallmark of adaptive radiations and a common signature of key innovations (22, 74–78, 90).

Suction cup-bearing fishes converge toward a broad, distinctive region of morphospace

Exceptional morphological diversity is another defining feature of key innovations and is often linked to adaptive radiation (73, 91, 92). If the pelvic suction cup functions as a key innovation, suction cup-bearing fishes should occupy a morphospace that is both expansive and distinct from that of nonsuction cup-bearing species. We generated a size-corrected mixed principal components analysis [PCAmix-data; (81)], which accommodates both continuous and categorical variables (Fig. 5A), allowing us to include both our continuous (compression/depression, elongation, body width, body depth, and head

depth) and categorical (scale presence, swim bladder presence, and tooth type) datasets. The first principal component (PC1, 49.1% of variance) represents lateral compression and dorsoventral depression, while the second (PC2, 19.6% of variance) reflects variation in body elongation. A phylogenetic PCA [phyl.pca in phytools; (60)] produced qualitatively similar results (figs. S11 to S14), confirming that the observed patterns were not biased by phylogenetic structure. In addition, to ensure that results were not driven by habitat affinity along the benthic-pelagic axis (79), we reran the analysis with only benthic taxa (figs. S15 to S18). In all analyses, the first two axes consistently represented the same morphological gradients (body compression/depression and body elongation, respectively). Phylogenetic correction reduced absolute variance along each axis while concentrating a greater proportion of the remaining variance in the first two axes. Removal of nonbenthic taxa decreased the amount of absolute and proportional variance explained by PC1, distributing variance more evenly across other axes (table S13). Benthic-only analyses recover the same qualitative separation of pelvic-fin states and the distinctiveness of suction cup-bearing taxa (figs. S15 to S18 and table S15).

Patterns of morphospace occupation differed markedly among pelvic-fin states. Species with separate pelvic fins were broadly distributed but concentrated in laterally compressed, deep-bodied regions of morphospace, consistent with morphologies typical of speciose reef lineages (93). United pelvic fins, restricted to gobies, clustered tightly near the center of morphospace, indicating an intermediate, near-cylindrical body form. Species lacking pelvic fins spanned a broad range of shapes but were most concentrated in elongate, eel-like regions. In contrast, suction cup-bearing species occupied a distinct region characterized by dorsoventrally depressed, stout-bodied forms, with limited overlap with other fin states (Fig. 5A). Prior studies have identified compression/depression and elongation as the dominant axes of body-shape variation in bony fishes (79, 94–97), yet the extreme depression end of this space is typically sparsely populated and separated by a gap (79, 94, 96). Suction cup-bearing fishes occupy this same region and expand it: They reach the extremes of monkfishes and batfishes, fill the gap, and extend more broadly along the elongation axis, altering the apparent structure of spiny-finned fish morphospace around moderate elongation.

A linear discriminant analysis showed that morphology significantly discriminates among pelvic-fin states (accuracy: 75.6%, $\kappa = 0.34$, $P < 2.2 \times 10^{-16}$). Classification accuracy was highest for suction cup-bearing species (balanced accuracy: 0.72), indicating that this condition is associated with the most distinctive multivariate body shape, whereas other pelvic states—particularly united (balanced accuracy: 0.50) and absent fins (balanced accuracy: 0.59)—exhibited greater morphological overlap with the separate-fin state (balanced accuracy: 0.65). These patterns were consistent across benthic-only and phylogenetically corrected analyses, with only modest declines in overall classification accuracy (see table S14).

Hypervolume analyses further supported the distinctiveness of suction cup morphology. Using hypervolume (98) and dynRB (99), we quantified hypervolume size, overlap, similarity, and distance. Suction cup-bearing species occupied the largest hypervolume across metrics (Fig. 5B and fig. S10) and showed low overlap and similarity with other states (Fig. 5, C and D, and fig. S9), driven mainly by separation along PC1 (Fig. 5A). Centroid distances between the suction cup and other states were also large (Fig. 5E), with the same qualitative pattern under Jaccard similarity and minimum-distance metrics

(fig. S8). A phylogenetically corrected PCA gave similar results, although suction cup hypervolume size was smaller than separate and absent states (figs. S11 to S14). Restricting to benthic taxa changed some pairwise comparisons, but suction cup morphospace remained distinct (low similarity/overlap and high distance) and was exceeded only by the absent state in overall size (figs. S15 to S18). Clade-level comparisons showed that suction cup-bearing clingfishes and cyclopteroids occupy larger, more differentiated hypervolumes than their closest relatives, whereas rock-climbing gobies occupy a smaller hypervolume with modest overlap (table S15 and fig. S19).

Last, because no single metric captures all aspects of convergent evolution (100), we applied three complementary approaches to test for convergence: Wheatsheaf index (101), Stayton's C/Ct statistics (100–102), and RRphylo (103) trajectory tests. The Wheatsheaf index indicated high but nonsignificant convergence ($w = 2.27$, $P = 0.51$). Stayton metrics, computed on five independent 25% stratified subsamples, gave modest C1 (0.155 to 0.178; significant in one subset) but larger C2 (0.621 to 0.891; significant in four subsets), while Ct metrics were not significant (table S16). RRphylo showed aligned evolutionary trajectories among the three suction cup-bearing lineages ($\theta < 90^\circ$; $P = 0.001$ to 0.019), strengthened by time-weighting ($\theta < 40^\circ$; $P = 0.001$; table S17). These results match the PCA: Suction cup-bearing taxa move toward a shared region of morphospace but remain spread across it, consistent with parallel shifts rather than tight convergence on a single body plan.

The impact of the suction cup on lineage diversification varies among suction cup-bearing clades

Although the role of accelerated lineage diversification rates as a necessary and sufficient criterion for identifying key innovations has been questioned in recent years (15), such shifts can still provide valuable supporting evidence (19, 65, 104, 105). We used the HiSSE framework (106) to estimate state-specific diversification rates, coding the pelvic suction cup as a binary trait (state 0: separate, united, or absent; state 1: suction cup). We conducted three independent analyses, one for each focal subclade, applying clade-specific sampling fractions. To further account for uneven sampling due to data deficiency within RCGobies+T1, we generated 10 reduced-taxon subsets with equal sampling fractions to maximize variability while maintaining taxonomic diversity (see Supplementary Data for the script and subsets) and pruned all 10 trees accordingly.

We fitted six models (character-independent and character-dependent with zero to two hidden states) to each subtree and compared them using AIC weights. Model selection (Fig. 6, E to G, insets) differed among subclades and was highly topology dependent in clingfishes+T1 (Fig. 6E). In balanced, subsampled RCGobies+T1 trees, a character-dependent BiSSE model was best in 95 of 100 replicates (table S18), whereas in the full RCGobies+T1 trees, a character-independent model with one hidden state was always favored; rate reconstructions nevertheless inferred a shift at the suction cup origin, although its direction varied across alternative trees. Most cyclopteroids+T1 trees favored HiSSE with two hidden states (AICw > 0.65 ; Fig. 6F, inset). These hidden regimes may reflect depth or latitude, which vary widely in cyclopteroids, especially snailfishes (39, 107), and can affect diversification (49, 108). Overall, diversification signals are clade dependent and topology sensitive, with the clearest suction cup-linked pattern in balanced RCGobies+T1 analyses.

Given this variability, we calculated model-averaged tip diversification rates between suction cup-bearing and cupless species for

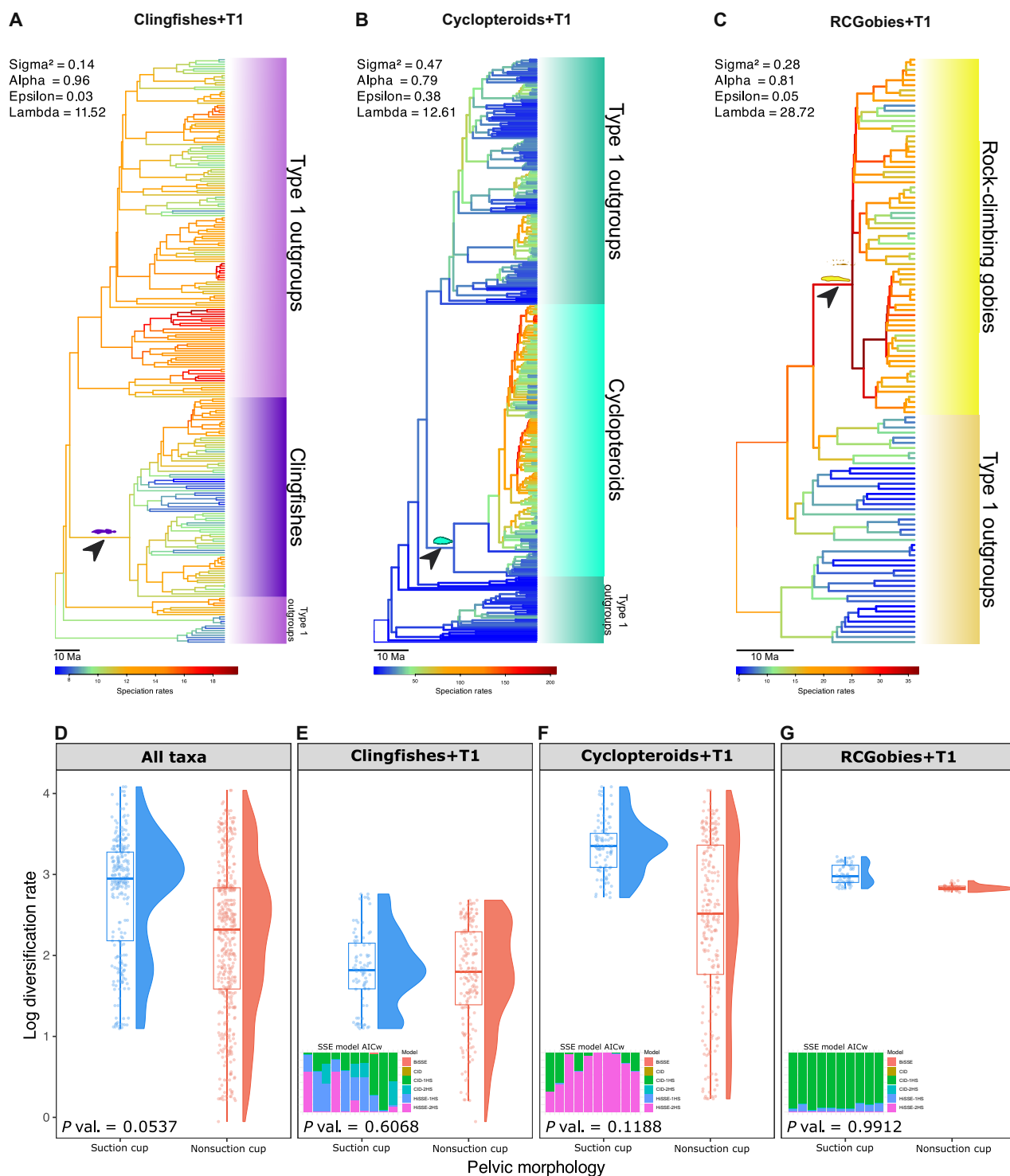


Fig. 6. Diversification rate analyses show overall faster rates of diversification in suction cup-bearing species. (A to C) CladeDS estimates mapped onto phylogenies for (A) clingfishes+T1, (B) cyclopteroids+T1 (snailfishes, lumpsuckers, and close relatives), and (C) RCGobies+T1; arrows mark suction cup gains. CladeDS parameters summarize rate dynamics: σ^2 (variance in speciation rates; highest in cyclopteroids+T1, lowest in clingfishes+T1), α (temporal trend; all three clades show declining rates), ϵ (turnover; highest in cyclopteroids+T1), and λ (initial speciation rate; highest in RCGobies+T1). (D to G) Tip-based diversification rates from HiSSE for the full tree and each subclade, with P values from phylogenetically corrected Wilcoxon tests. Inset stacked bar plots represent the AIC weights for each state specific diversification and extinction (SSE) model for each tree.

each subclade and (as a pooled tip-rate comparison) the T2-pruned tree and then used a phylogenetically corrected Wilcoxon test (109) as the data did not meet normality assumptions. When all taxa were analyzed together (single T2-pruned tree), suction cup-bearing species exhibited higher rates, with a marginally significant difference ($P = 0.054$; Fig. 6D). In the three subclades, there were no significant differences in net diversification rates between suction cup-bearing and nonsuction cup-bearing species. The violin plots of phylogenetically corrected tip rates (Fig. 6, E to G) suggest higher rates in suction cup-bearing lineages, and the stronger signal when pooling subclades suggests that the subclade-specific tests may be limited by small sample sizes and the lower power of nonparametric tests (110).

Last, to test for bursts of diversification following the origin of the suction cup within each subclade, we estimated branch-specific diversification rates using ClaDS analyses (111, 112). Diversification rate patterns varied across subclades (Fig. 6, A to C). In clingfishes+T1 (Fig. 6A), net diversification rates were relatively uniform and showed no detectable increase with the origin of the suction cup. Nevertheless, clingfishes exhibited rates comparable to those of blennies, one of the fastest-radiating teleost lineages (29, 113), whose exceptional diversity is attributed to repeated transitions to cryptobenthic reef habitats, possibly in part because clingfishes have also shifted into cryptobenthic niches (114). In cyclopteroids+T1 (Fig. 6B), diversification rates were highly variable, with low rates at deeper nodes and some accelerations throughout the clade, but with the most dramatic rate shifts occurring within snailfishes. Although a moderate acceleration was estimated at the root of lumpsuckers, no clear shift coincided with the suction cup in the common ancestor of the families. RCGobies+T1 (Fig. 6C) showed a pronounced burst of diversification associated with the evolution of the suction cup before tapering off but remaining above pre-cup levels, aligning with the results of the subsampled HiSSE analysis. Together, these patterns suggest that the pelvic suction cup can promote diversification, but its effects are contingent on clade-specific evolutionary contexts.

DISCUSSION

Our dataset provides an integrative test of the pelvic suction cup as a key innovation in spiny-finned fishes, revealing consistent macroevolutionary signatures across three independent origins spanning over 100 MY. Across our time-calibrated 960-species phylogeny, suction cup-bearing lineages exhibit evolutionary asymmetry with repeated losses but no regains or reversions to separate pelvic fins, accelerated phenotypic evolution around unique trait optima consistent with a wide and depressed body plan, and expansion into a broad and previously underexplored region of morphospace. Suction cup-bearing fishes occupy and expand the sparsely populated, extremely depressed end of percomorph morphospace, converging toward a functional body plan defined by strong dorsoventral depression and moderate elongation that increases adhesive surface area and reduces drag in high-energy environments while still diversifying within that space. The suction cup is also associated with multicuspoid teeth, which, in rock-climbing gobies, correlate with herbivory, consistent with access to algae-rich substrates.

The evolutionary consequences of the suction cup, however, are not uniform across lineages. In rock-climbing gobies, the suction cup is associated with reduced ecofunctional differentiation, consistent with specialization following the transition to fast-flowing stream habitats, where the suction cup enables them to scale waterfalls and

access upstream reaches unavailable to other fishes. Despite their limited morphological divergence, which may reflect constraint on a generalized goby body shape or preadaptation for life in swift currents, rock-climbing gobies show the clearest diversification burst coincident with suction cup origin. Facultative amphidromy, often associated with accelerated diversification (115), evolved before the suction cup in gobies and became obligate only in rock-climbing gobies (47, 116) and, therefore, does not solely account for the diversification shift. This pattern points to the suction cup as a key evolutionary catalyst even without accelerated morphological divergence. In clingfishes, the suction cup is linked to higher morphological rates around adaptive optima and marginal increases in ecofunctional differentiation consistent with broader microhabitat use, with their powerful adhesive disc allowing them to occupy high-energy tidal habitats inaccessible to other fishes. Suction cup-bearing cyclopteroids explore a broad range of morphospace around optima, yet we detected no shift in ecological or functional space occupancy relative to closely related outgroups, possibly because data gaps limited our ability to code key traits across species in the clade (e.g., parasitic egg deposition) despite the breadth and distribution of snailfishes, which span tidepools to the hadal zone and exploit habitats ranging from wave-swept intertidal rocks to deep-sea crab gills.

Although our results support repeated origins of the pelvic suction cup with clear macroevolutionary consequences, several limitations constrain interpretation. These limitations underscore the need for nuance in interpreting and identifying key innovations as their macroevolutionary consequences are not uniform or deterministic but instead depend on lineage-specific ecological contexts, trait interactions, and underlying developmental architectures. In addition, phylogenetic sampling remains sparse for many snailfish and goby species, limiting transition-rate estimates in cyclopteroids, where ecological data are also lacking, and creating imbalanced sampling fractions in gobies. Small sample sizes in tip-rate analyses also reduce power to detect finer-scale diversification shifts. Functional and morphological data for the pelvic girdle and disc in gobies are likewise limited, so additional transformations with functional repercussions may have been overlooked. Expanded taxonomic sampling, together with detailed morphological and functional analyses, will be needed to refine transition estimates and clarify the full range of structural modifications associated with suction cup evolution. At the genomic scale, although the suction cup represents many-to-one mapping, repeated origins may still involve shared developmental or regulatory mechanisms, as suggested by whole-genome comparisons of repeated paired-fin reduction in percomorphs (117). Related comparative frameworks could also be applied to other adhesive innovations in teleosts (e.g., loricariid oral discs, remora dorsal discs, and hillstream loach ventral discs) to test for parallel functional and genomic patterns across phylogenetically distant lineages and link macroevolutionary patterns to the mechanisms enabling repeated origins over deep evolutionary timescales.

In sum, our results across three distantly related spiny-finned fish lineages show that the suction cup reshaped morphospace, accelerated phenotypic evolution, and in some clades increased lineage diversification. By enabling attachment to and exploitation of high-energy benthic habitats, the suction cup demonstrates how a single morphological novelty can repeatedly transform evolutionary trajectories. Wainwright and Longo (30) identified eight functional innovations contributing to adaptive radiations in spiny-rayed fishes. The consistent signatures of innovation demonstrated here firmly establish

the pelvic suction cup as an additional key innovation in the evolution of spiny-rayed fishes.

MATERIALS AND METHODS

Extended Materials and Methods are provided in the online Supplementary Materials, Extended Materials and Methods. All data and R scripts used for analyses are available in the Figshare repository (DOI:10.6084/m9.figshare.30593363).

Genomic data assembly and alignment

We assembled a dataset of 960 percomorph species designed to test the independent origins of the pelvic suction cup, with dense sampling of rock-climbing gobies, clingfishes, and cyclopteroids and their close relatives (sampling fractions: 0.32 for cyclopteroids and 0.35 for their T1 outgroups, 0.45 for clingfishes and 0.15 for their T1 outgroups, and 0.57 for rock-climbing gobies and 0.09 for their T1 outgroups), plus broad percomorph representation (T2 outgroups). Data came from newly generated exon capture libraries, published FishLife exons, SRA accessions, genome assemblies, and legacy markers, including 67 newly shotgun-sequenced snailfish genomes. Exons were harvested using FishLife (118) and BUSCO pipelines (53), filtered for quality, and aligned with MACSE (119). After excluding poor alignments and sparsely sampled species, the final genomic dataset comprised 940 FishLife exons and 1960 BUSCO genes across 616 species.

Phylogenetic reconstruction, time calibration, and subclade grafting

We inferred phylogenetic relationships using both concatenation-based ML and multispecies coalescent approaches, estimating snailfish trees separately, using both FishLife and BUSCO genes. To account for phylogenetic uncertainty, we built four gene subsets optimized for taxonomic coverage (120). With the complete dataset and four subsets, we generated five ML trees with IQ-TREE (121) and gene trees with RAXML-NG (122) to infer five species trees with ASTRAL-IV (123). Additional taxa (344 species) were added through backbone constraint analyses with nine legacy markers, yielding expanded percomorph trees. Snailfish-only trees based on assembled genomes were inferred from FishLife markers and BUSCO genes. All trees were time-calibrated using 246 secondary calibrations in TreePL (124), and snailfish trees were subsequently grafted onto the expanded percomorph backbones, replacing the smaller subset of snailfishes on the initial backbones, to produce 10 calibrated trees (See Supplementary Data for all 10 topologies and for calibration file).

Fitting of custom transition models of pelvic morphology and ancestral state reconstructions

We coded pelvic-fin morphology into four discrete states (0: separate, 1: united, 2: suction disc, and 3: absent). State 1 was restricted to gobies with fully united pelvic fins forming a round disc (125) but where disc function is unknown or cannot be inferred from pelvic skeletal morphology. Only rock-climbing gobies were coded as having a true suction cup (state 2). Although some other fishes [i.e., hillstream loaches; (126, 127)] have pelvic fins that have been modified to facilitate suction, we defined the pelvic suction cup as requiring both fused pelvic fins and skeletal modifications of the pelvic girdle that together enable the generation of strong suction force—that is, both form and function must be present. We fit 10 models of

discrete trait evolution: three standard Mk models (ER, ARD, and SYM), their threshold counterparts (ERu, ARDu, and SYMu), and four custom models (OLI, ordered loss irreversible; OLR, ordered loss reversible; ULI, unordered loss irreversible; and ULR, unordered loss reversible). The custom models varied in whether transitions were ordered or unordered and whether fin loss was reversible, but all prohibited reevolution of the suction cup once lost. Transition rate models with hidden rate categories or unobserved states were not used when testing our biologically informed custom models because functions to fit these models [phytools (60) and corHMM (61)] do not allow the use of custom, biologically informed transition matrices. Using a root-state prior that fixed the root at state 0 according to our biologically informed hypothesis, we fit these models across the full percomorph tree and each suction cup-bearing subclade [rock climbing gobies and their closest relatives (RCGobies+T1); snailfishes, lumpsuckers, and their closest relatives (cyclopteroids+T1); and clingfishes and their closest relatives (clingfishes+T1)] using fitMk in phytools (60), compared models using likelihood ratio tests and AIC weights, and reconstructed ancestral states using model averaging with both marginal (full tree; Fig. 2 and fig. S1, function anc in phytools) and stochastic mapping on the snailfish-only tree shown in the Supplementary Materials (fig. S3; make.simmap in phytools), with the root-state prior fixed at state 2 based on the full-tree marginal reconstructions. To assess the robustness of our estimates of suction cup origins, we additionally performed parsimony-based ancestral state reconstructions on all 10 trees using castor (128) function asr_max_parsimony, which performs well when the rate of evolution for a trait is slow. Last, to validate rate estimates, we fit ARD continuous-time Mk models using corHMM, including hidden-state formulations by evaluating rate-category models with zero, one, or two hidden rate categories with fixed root-state priors at state 0. For each tree, model support was compared by AICc, and the best-supported model was retained for downstream transition rate estimates, extracted from the fitted Q matrices (table S5).

Assessing correlations between the pelvic suction cup and discrete and continuous characters

We tested correlations between suction cup presence and three discrete traits: tooth type (538 spp.), scale presence (940 spp.), and swim bladder presence (929 spp.), all scored from the literature (see Supplementary Data). Following (65), we coded teeth as simple (unicuspid) or complex (multicuspid); we scored scales and swim bladders as present or absent in adults. For continuous traits, we measured elongation, depression, body width, body depth, and head depth from 429 specimens in the SIO Marine Vertebrate Collection and combined them with 1070 records from the FishShapes v1 database (66), yielding size-corrected values for 1499 individuals across 538 species. We standardized continuous variables by dividing raw measurements by a composite body-size metric (the cube root of standard length $\times$ maximum fish width $\times$ maximum body depth) and log-transforming (see Supplementary Methods for details on measurement technique and calculation).

We tested correlations between discrete traits using hidden-state correlation models in corHMM (61) with complementary phylogenetic regression in Phylolm (67). For each suction cup–trait pair, we compared independent and dependent models of binary trait evolution, each with and without hidden-rate categories, to distinguish correlated evolution from background rate heterogeneity. We also evaluated multiple root treatments and selected the best fit by AIC (see Supplementary Methods). We tested correlations between discrete

and continuous traits with phylogenetic linear regression (PhyloIm), treating the suction cup as either predictor (function phyloIm) or response (function phyloglm). For phyloIm, we fitted six evolutionary covariance models on all 10 time-calibrated trees for the full phylogeny and each subclade: BM, OU with either random (OUrandomRoot) or fixed (OUfixedRoot) roots (to account for uncertainty in ancestral trait distributions), EB, and Pagel's λ , κ , and δ . We compared model fits with AICc, summarized correlation strength with R^2 values and odds ratios (interpreted as fold changes in traits when the suction cup was the predictor, or multiplicative changes in the odds of suction cup presence when it was the response) and assessed significance using Benjamini-Hochberg FDR corrections applied within traits and subclades.

Fitting models of continuous trait evolution

To test whether rates and modes of continuous trait evolution varied with pelvic-fin morphology, we fit four models: single-rate BM (BM1), multirate BM (BMS), single-regime OU (OU1), and multiregime, multirate OU (OUMV). We assigned regimes according to pelvic-fin state (separate, united, suction cup, absent) and mapped them onto the phylogeny. We tested all models in OUwie (80) across replicate trees for the full phylogeny and each subclade. We compared models using AICc and AICw, excluding models that produced negative eigenvalues and unrealistically high log-likelihoods, which only included variable α (strength of selection) models (OUMA and OUMVA), and summarized support across replicates to identify the best-fitting model for each trait. For traits that received mixed support (maximum AICw < 0.95), we estimated rates based on model averaging (table S9).

Morphospace and convergence analyses

We constructed a morphospace using a mixed PCA [PCAmix, R package PCAmixdata (129)] to jointly analyze continuous body-shape and categorical traits. We standardized continuous variables by body size and encoded categorical traits as descriptive factors. On the basis of the Cattell scree plot of the eigenvalues (see fig. S5), we retained the first four PCs, which together explained 88.9% of the variation, and plotted individual species colored by pelvic-fin state. To quantify the extent to which morphology differentiates pelvic-fin states, we performed a linear discriminant analysis (130) using PC1 to PC4 as predictors and pelvic-fin state as the grouping variable, evaluating classification performance with repeated 10-fold cross-validation (50 repeats). To incorporate phylogenetic information, we performed a phylogenetic PCA [phyl.pca under a BM model; phytools; (60)] on continuous traits, retaining the first four PCs and projecting tree topologies into phylomorphospace (fig. S11). We quantified morphospace occupation by pelvic-fin state using four-dimensional hypervolumes based on PCs 1 to 4 [hypervolume package: hypervolume_box, hypervolume_set; (98)]. These methods estimate the multivariate trait-space occupied by each group from the point cloud, allowing direct comparison of hypervolume size, overlap (Jaccard and Sørensen), centroid distances, and minimum intergroup distances. As a complementary nonparametric approach, we used dynRB [dynRB_VPa; (99)], which quantifies overlap and dispersion directly from the spatial arrangement of points in multivariate space (and does not require specifying an explicit parametric model). We visualized these comparisons as heatmaps and barplots. Agreement between the two approaches supports that the morphospace differences reflect consistent shifts in multivariate trait distributions rather than dependence on a single method's assumptions.

We quantified phenotypic convergence among suction cup-bearing lineages using three complementary frameworks applied to morphospace PC scores. We estimated phenotypic clustering with the Wheat-sheaf index (101), assessed convergence through time using Stayton's C and Ct metrics (102), and tested parallelism in evolutionary trajectories using RRphylo (103). Statistical support for each approach was evaluated against null expectations generated by simulation on the corresponding phylogeny; details on simulations, subsampling, and sensitivity analyses are provided in Supplementary Methods.

Ecological diversification and functional distinctiveness/specialization analyses

To characterize species' functional profiles, we compiled ecological and life-history data from primary literature and scientific databases (see Supplemental Data). We assembled clade-specific traits reflecting relevant axes of ecological diversity within each subclade (albeit constrained by data availability). For clingfishes+T1, we tested substrate preference (biogenic, hard, mixed, and soft) and water ecotype (deep, shallow, intertidal, and fluvial). For cyclopteroids+T1, we tested depth range following (131), ecoregion following (49, 132), and water column position (pelagic or nonpelagic). For RCGobies+T1, we tested trophic guild (generalized carnivore, herbivore/detrivore, mobile invertivore, omnivore, planktivore, and sessile invertivore) and water ecotype (marine, marine and freshwater, coastal freshwater, and above waterfalls). See table S11 and Supplementary Data. We quantified functional differences using a Gower distance matrix [function gowdis, R package FD (89)] to accommodate mixed categorical and continuous traits. Functional distinctiveness was calculated as the mean pairwise distance among species (133). Functional specialization was estimated by conducting a PCoA with a Lingoes correction and measuring each species' distance to the multivariate centroid across all positive axes. To test whether suction cup presence influenced functional profiles, we fitted phylogenetic generalized least-squares models using the R package caper (134), comparing them to ordinary least squares models. We constrained Pagel's λ to near-zero and near-one bounds to contrast nonphylogenetic and phylogenetic models. Analyses were repeated using both a multispecies coalescence and maximum likelihood topology (full exon sets) to account for phylogenetic uncertainty. To elucidate potential evolutionary correlations between tooth type and ecological traits, we performed multivariate and univariate phylogenetic generalized logistic regression analyses, described in (135), and implemented in R package phyloIm (67). Analyses were conducted independently on the three subclades on all 10 trees with 1000 bootstrap replicates. All *P* values were corrected via Benjamini-Hochberg FDR.

State-dependent diversification and SSE model testing

We assessed whether the suction cup influenced diversification using state-dependent speciation-extinction (SSE) models fit with HiSSE (106) for each suction cup-bearing subclade separately, excluding the full percomorph trees to avoid biases from uneven outgroup sampling. We modeled the suction cup as a binary trait and fit six models: character-independent with no hidden states, BiSSE, character-independent with one hidden state, HiSSE with one hidden state, character-independent with two hidden states, and HiSSE with two hidden states. We compared models using AIC, but results were often variable and inconclusive. To obtain robust estimates, we averaged tip diversification rates estimated from all HiSSE models with AIC weight >0.05 for each of the 10 trees per subclade and

then compared rates between suction cup-bearing and nonsuction cup-bearing taxa with phylogenetically corrected Wilcoxon tests. Last, we used ClaDS (111, 112) to reconstruct lineage-specific temporal patterns of diversification.

Supplementary Materials

This PDF file includes:

Supplementary Methods

Tables S1 to S18

Figs. S1 to S22

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Acknowledgments: We thank K. Maslenikov (UW Burke Museum), B. Frable (Scripps Institution of Oceanography), B. Victor, S. Kortet, the Te Papa Museum of New Zealand, and J. Gardner (UW) for assistance in locating and contributing specimens to this project. Computational support was provided by the University of Oklahoma Supercomputing Center for Education & Research (OSCER). **Funding:** This work received funding from the National Science Foundation grant DEB-2225130 (R.B.-R.), National Science Foundation grant DEB-1541491 (R.B.-R.), and National Science Foundation grant DEB-1906574 (E.C.S.). **Author contributions:** Writing—review and editing: E.P.M., A.P.M.M., M.M., G.O., L.C.H., C.C.B., E.C.S., J.W.O., D.A., L.M.T., and R.B.-R. Resources: L.C.H., C.D.R.P.-M., M.M., S.P., W.T.W., G.O., C.C.B., A.S., E.C.S., J.W.O., T.D.L., D.A., L.M.T., and R.B.-R. Data curation: E.P.M., W.T.W., L.C.H., E.C.S., and R.B.-R. Investigation: E.P.M., A.P.M.M., L.M.T., and D.A. Formal analysis: E.P.M. and A.P.M.M.

Conceptualization: E.P.M., G.O., E.C.S., D.A., L.M.T., and R.B.-R. Methodology: E.P.M., D.A., L.M.T., and R.B.-R. Funding acquisition: S.P., D.A., and R.B.-R. Validation: E.P.M., J.W.O., L.M.T., and D.A. Supervision: R.B.-R. Project administration: R.B.-R. Writing—original draft: E.P.M. and R.B.-R. Software: E.P.M. Visualization: E.P.M. **Competing interests:** The authors declare that they have no competing interests. **Data, code, and materials availability:** All data and code needed to evaluate and reproduce the results in the paper are present in the paper, the Supplementary Materials, or the associated Figshare repository (DOI:10.6084/m9.figshare.30593363). Newly generated genomes are available at the National Center for Biotechnology Information Sequence Read Archive BioProject (numbers PRJNA1198512–PRJNA1198578).

Submitted 6 April 2026

Accepted 21 July 2026

Published 4 September 2026

10.1126/sciadv.aeh7851

The convergent evolution of the pelvic suction cup: A unique key innovation in spiny-finned fishes

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Sci. Adv. **12** (36), eaeh7851. DOI: 10.1126/sciadv.aeh7851

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